Time-Series Foundation Models as Approximate-Information-State World Models:

A Fisher-Information Theory of Stability, Observability, Learnability, and Chaotic Predictability

(fusing the world-action / vision–language–action Fisher theory with the CHRONOS time-series foundation model, on the approximate-information-state framework of Subramanian, Sinha, Seraj & Mahajan, *JMLR* 2022)

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Abstract

A time-series foundation model (TSFM) such as CHRONOS is trained to predict the next value of a sequence by tokenization and cross-entropy—the language model recipe transplanted to time series. A world-action model (WAM) is trained to predict an environment's next observation given actions. We show these are the *same object with the action channel deleted*: a TSFM parameterizes the environment (observation) channel of a partially observed process on a shared approximate-information-state (AIS) latent, its transformer context *is* the AIS, and its statistical efficiency is governed by *one* Fisher information matrix (FIM)—the innovations / system-identification block $\mathcal{I}_{\theta\theta}$ of the WAM/VLA theory, now standing alone because a forecaster takes no actions (the policy block and the occupancy coupling vanish). This single identity carries the entire mathematical apparatus of the two-block theory into forecasting: the tangent-filter FIM, the small-eigenvalue diagnostic, and the data-versus-sensor dichotomy. Our central contribution is to show that, for the linear–Gaussian (LQG) instance, the FIM's eigenstructure is dictated by two physical quantities: *stability margins* and *observability*. We prove a stability–identifiability law—the per-mode information obeys the AR(1) form $\Gamma_{jj} \asymp \mathrm{gain}_j^2\, \sigma_w^2/(\sigma_v^2(1-r_j^2))$, so $\lambda_{\max}(\mathcal{I}_{\theta\theta}) \asymp 1/(1-\rho(A))$ diverges and the FIM condition number blows up as the spectral radius $\rho(A) \to 1$—formalizing a tension between *learnability* (information grows near the unit circle) and *conditioning* (the sloppy spectrum widens). The eigenvectors of the small eigenvalues are exactly the *fast* modes (excitation/data-limited, curable by longer series) and the *unobserved* modes (sensor-limited, curable only by a new observation channel), so the WAM/VLA data-versus-sensor dichotomy becomes a statement about which time series a foundation model can and cannot learn to forecast. A large numerical study on a stable LQG system with 200 latent states and 100 observations confirms every claim: the measured FIM has exactly as many null directions as unobserved modes, its per-mode information tracks $1/(1-r^2)$ to a log–log correlation of 0.97, $\lambda_{\max}$ scales as $1/(1-\rho)$, and a trained CHRONOS-lite model's forecast skill rises with stability and tracks $\log\det \mathcal{I}_{\theta\theta}$, while sloppy directions are learned last in exact proportion to their Fisher eigenvalue. Beyond the linear regime we develop the mathematics of forecasting *chaotic* systems and route it through the same information-state latent. On a corpus of five dynamical systems—a stable modal LQG process and four nonlinear oscillator/field models (multi-population Kuramoto, cluster Stuart–Landau, Kuramoto–Sivashinsky, and Lorenz–96)—we prove that the transformer context is a *delay-embedding* approximate information state whose intrinsic dimension is the *attractor's*, not the ambient n (Takens), which is why a modest-context model forecasts a nominally 200-dimensional system; that the forecast horizon of *any* model is capped at the Lyapunov time $\tau_\star \asymp \lambda_1^{-1}\log(1/\sigma_v)$; that a filter/AIS tracks the state with bounded error iff it observes the unstable subspace, so a low-dimensional sensor on a high-dimensional chaos (we measure $n_+ = 13$ positive exponents and Kaplan–Yorke dimension $D_{\mathrm{KY}} \approx 27$ for Lorenz–96) is *dynamically* sensor-limited; and that the Kolmogorov–Sinai entropy $h_{\mathrm{KS}} = \sum_{\lambda_i>0}\lambda_i$ lower-bounds the tokenizer information rate $m\log B$ (Pesin). This places the linear theory as one pole of a single axis: the top Lyapunov exponent $\lambda_1 = \log\rho$—which our five systems sweep from -0.105 (LQG, contracting) through ≈ 0 (synchronized oscillators) to $+1.68$ (Lorenz–96)—is the master dial, and the parameter-information blow-up $1/(1-r^2)$ and the predictability horizon $1/\lambda_1$ are the *same* $1/|\lambda_1|$ singularity approached from the stable and chaotic sides of the unit circle. The common theme: world modeling, instruction-conditioned control, and time-series forecasting are three channels of one process measured by one Fisher matrix whose spectrum is written by dynamics.

Key words. time-series foundation models, world models, Fisher information, approximate information state, system identification, stability margin, observability, Cramér–Rao bound, sloppy models, Chronos, chaotic dynamics, Lyapunov exponents, Takens delay embedding, predictability horizon, Kolmogorov–Sinai entropy, data assimilation, Kuramoto–Sivashinsky, Lorenz–96

Table 1: Three foundation-model paradigms as *channels of one partially observed process* on a shared approximate-information-state (AIS) latent. A WAM predicts an environment's next *image* given the action taken; a TSFM predicts a numerical series' next value with *no* action; a VLA predicts the next *action* from an image and a language instruction. They differ only in which modality they observe and in whether they emit or condition on actions—and hence in which block of the *one* Fisher information matrix governs their learning. Deleting the action channel collapses the joint FIM to the single system-identification block $\mathcal{I}_{\theta\theta}$ shared by WAM and TSFM (Eq. (1)).

	World–Action Model (WAM)	**Time-Series Foundation** (TSFM)	**Vision–Language –Action** (VLA)
Observation o_t	images / video frames (pixels)	numerical values (scalar or vector)	images (+ proprioception)
Conditioning input	past frames *and* action a_t	past values only	image *and* language instruction ℓ
Emits (target)	next observation o_{t+1}	next value o_{t+1}	next action a_t
Takes actions?	conditions on them	none (autonomous)	*produces* them
Language / instruction?	no	no	yes (ℓ)
Channel of the process	world $P_\theta(o \mid h^-, a)$	world $P_\theta(o \mid h^-)$	policy $\pi_\phi(a \mid h, \ell)$
Parameter learned	world θ	world θ	policy ϕ
Governing Fisher block	$\mathcal{I}_{\theta\theta}$	$\mathcal{I}_{\theta\theta}$ (*sole* block)	$\mathcal{I}_{\phi\phi}$
Representative models	Genie, GAIA-1, DreamerV3	CHRONOS, TimesFM, Moirai	RT-2, OpenVLA, π_0

AMS subject classifications. 62M10, 62B10, 93E12, 93B07, 93B30, 68T07, 62M20, 93E11, 37M25, 37D45, 37A35, 37N30, 60G25

1 Introduction

Two research programs that rarely cite each other have arrived at the same mathematical place. On one side, *world models* and *vision–language–action* (VLA) policies learn, respectively, to predict an environment's future and to act in it; a companion paper [1] shows that these are the two factors of one partially observed process on a shared approximate-information-state (AIS) latent [2], and that their learning problems are the two orthogonal blocks of a single Fisher information matrix (FIM). On the other side, *time-series foundation models* (TSFM) such as CHRONOS [3] forecast an arbitrary numerical sequence by *tokenizing* its values and training an off-the-shelf language model with the cross-entropy loss—"the language of time series." These look like unrelated enterprises: one is embodied, generative, and control-theoretic; the other is a minimalist repurposing of a text transformer for forecasting. This paper shows they are one enterprise (Table 1).

The thesis in one sentence. A time-series foundation model is an *autonomous world-action model*: it parameterizes the observation channel $P_\theta(o_t \mid h_t^-)$ of a partially observed dynamical process, with the action channel deleted because a forecaster does not act; its context embedding is an approximate information state; and its statistical efficiency is governed by the *single* Fisher block that survives when the policy block is removed—the innovations, system-identification FIM. Everything the WAM/VLA theory says about that block [1]—the tangent-filter computation, the eigenstructure of identifiability, the small-eigenvalue diagnostic, the data-versus-sensor dichotomy—transfers verbatim to forecasting, and acquires there a sharp new reading in terms of the two quantities a control theorist cares about most: *stability* and *observability*.

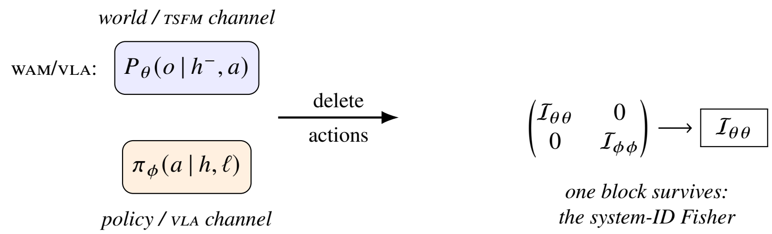

Figure 1: The fusion (1). A time-series foundation model is a world-action model with the action channel deleted: the policy block $\mathcal{I}_{\phi\phi}$ and the cross block $\mathcal{I}_{\theta\phi}$ vanish, and the joint Fisher information collapses to the single world (system-identification) block $\mathcal{I}_{\theta\theta}$. The TSFM is the single-block corner of the two-block theory of [1].

Why the fusion is exact, not analogical. Figure 1 is the whole idea. The WAM/VLA trajectory law factors into an environment channel (emitting observations, parameterized by the world θ) and a policy channel (emitting actions, parameterized by the VLA ϕ), and the joint FIM is block-diagonal, $\mathcal{I} = \operatorname{blkdiag}(\mathcal{I}_{\theta\theta}, \mathcal{I}_{\phi\phi})$ [1, Thm. 6.1]. A TSFM forecasts an *uncontrolled* sequence: there are no actions, so the policy factor is absent, ϕ does not exist, the block $\mathcal{I}_{\phi\phi}$ and the cross block $\mathcal{I}_{\theta\phi}$ vanish identically, and the joint FIM collapses to its *world* block alone,

$$\mathcal{I}(\theta, \phi) = \begin{pmatrix} \mathcal{I}_{\theta\theta} & 0 \\ 0 & \mathcal{I}_{\phi\phi} \end{pmatrix} \xrightarrow[\text{delete actions}]{\text{TSFM}} \mathcal{I}(\theta) = \mathcal{I}_{\theta\theta}. \tag{1}$$

A TSFM is thus the pure-θ, single-block corner of the two-block theory: a world model with nothing to control. This is not a metaphor—it is (1), and it means the TSFM's information geometry is a *special case* of the embodied one, obtained by setting a channel to null. The payoff is that the difficult, control-theoretic half of the WAM/VLA theory—the world block $\mathcal{I}_{\theta\theta}$, which is the FIM of *system identification*—is exactly what a TSFM optimizes, and it is far richer than the forecasting literature's usual error metrics suggest.

The common mathematical theme. Once a TSFM's learning problem is identified with system identification through $\mathcal{I}_{\theta\theta}$, a single principle organizes all three paradigms—world modeling, instruction-conditioned control, and time-series forecasting:

> *Each model estimates a channel of one partially observed process on a shared AIS latent; its statistical efficiency is one Fisher information matrix; and the eigenstructure of that matrix is written by the system's* dynamics—*the stability margins set the* magnitude and conditioning *of information, and the observability sets its* null space. *The small Fisher eigenvalues are exactly the fast and the unobserved modes—the directions no amount of the wrong resource can learn.*

For a forecaster this theme is unusually concrete, because for a linear–Gaussian system the FIM has a closed form and its spectrum is *legible*: each eigenvalue belongs to a physical mode, and its size is the mode's stability margin weighted by its observability. That legibility is what this paper exploits.

Contributions.

1. **The identity (Section 2).** We cast a TSFM as the observation channel of a partially observed process on an AIS latent, show its FIM is the single world block $\mathcal{I}_{\theta\theta}$ of [1] (Proposition 2.3), and place CHRONOS's tokenization inside the theory as a quantized observation channel whose FIM is data-processing dominated by the continuous one (Proposition 2.2).
2. **The context is the AIS (Section 3).** We show the TSFM's context embedding is an (ε, δ)-AIS and transport the AIS Fisher bound of [1] to forecasting (Proposition 3.1).
3. **Stability writes the spectrum (Section 4).** We prove the stability–identifiability law: per-mode information obeys the AR(1) form $\Gamma_{jj} \asymp \mathrm{gain}_j^2/(1 - r_j^2)$, so $\lambda_{\max}(\mathcal{I}_{\theta\theta}) \asymp 1/(1 - \rho(A))$ and the FIM condition number diverges as $\rho(A) \to 1$ (Theorem 4.1); learnability and conditioning are in tension.

4. **Small eigenvalues, forecasting edition (Section 5).** We specialize the data-versus-sensor dichotomy: a sloppy Fisher direction is either a fast but observed mode (curable by more data) or an unobserved mode (curable only by a new sensor/covariate), and the fat observation map $m < n$ of a realistic TSFM *guarantees* a sensor-limited null space (Theorem 5.1), an observability ceiling on univariate forecasting.

5. **Learnability (Section 6).** We tie the FIM spectrum to learning curves through the Cramér–Rao bound: the excess forecast risk along a mode decays as $1/(T\Gamma_{jj})$, so slow modes are learned first and sloppy modes last, and aggregate skill tracks $\log\det \mathcal{I}_{\theta\theta}$ (Proposition 6.1).

6. **When the second block returns (Section 7).** Covariate/exogenous-input forecasting re-introduces an action-like channel and with it the policy block and the occupancy coupling of [1], locating covariate TSFMs back in the full two-block theory (Proposition 7.1).

7. **The model zoo, made explicit (Section 8).** We document the generative corpus—a stable modal LQG process and four nonlinear oscillator/field systems (multi-population Kuramoto, cluster Stuart–Landau, Kuramoto–Sivashinsky, Lorenz–96)—giving each system's exact transition structure and every hyperparameter, alongside the CHRONOS-lite architecture and training recipe, so the theory below is anchored to reproducible objects (Tables 2 to 4).

8. **Chaotic predictability and the information state (Section 9).** We extend the theory past the linear–Gaussian regime. We prove that a TSFM's context is a *delay-embedding* AIS whose dimension is the attractor's (Takens; Theorem 9.2); that the forecast horizon of any model is the Lyapunov time $\tau_\star \asymp \lambda_1^{-1}\log(1/\sigma_v)$ (Theorem 9.4); that bounded state tracking requires observing the unstable subspace, giving a *dynamic* observability ceiling (Theorem 9.6); that $h_{\mathrm{KS}} = \sum_{\lambda_i>0}\lambda_i$ lower-bounds the tokenizer rate $m\log B$ (Pesin; Proposition 9.8); and that the whole paper is one axis in the top Lyapunov exponent, with $1/(1-r^2)$ and $1/\lambda_1$ the two sides of the $\lambda_1 \to 0$ singularity (Proposition 9.10). We measure the Lyapunov spectra of the corpus to instantiate every constant.

9. **A large numerical study (Section 10).** On a stable LQG system with $n = 200$ states and $m = 100$ observations we confirm all of the above, including a trained CHRONOS-lite whose forecast skill is predicted by the FIM.

Related work. CHRONOS [3] and the concurrent TSFMS [10, 11, 12, 13] forecast by pretraining sequence models on large time-series corpora; CHRONOS specifically tokenizes by scaling and quantization and trains with cross-entropy ("regression via classification"). The idea that a next-token predictor is a maximum-likelihood system identifier is classical in the innovations / prediction-error framework [4, 5]; we make the identification explicit and route it through the AIS latent and the two-block FIM of [1]. The stability–identifiability law generalizes the AR(1) folk fact $I(a) = 1/(1-a^2)$ and the near-unit-root asymptotics of Phillips [6]; the "sloppy" spectrum spanning many decades is the time-series instance of sloppy-model theory [7, 8]. Observability and the parameter-sensitivity Gramian are standard in identification [4, 9]; our contribution is to read the small Fisher eigenvectors of a *foundation model* through them. The chaotic half of the paper rests on classical dynamical-systems theory: the delay-embedding theorems of Takens [14] and Sauer–Yorke–Casdagli [15], the multiplicative ergodic theorem of Oseledets [16], Pesin's entropy formula and its SRB extension [17, 18, 19], the Kaplan–Yorke dimension [20], and the predictability limits of chaotic forecasting [21, 22]. That a filter need only control the unstable subspace is the "assimilation in the unstable subspace" principle of Trevisan–Palatella and Bocquet–Carrassi–Grudzien and co-authors [23, 24, 25]; recent learning-theoretic work links forecasting skill on chaotic systems to the Lyapunov time [26]. We connect this apparatus to the *information state* of a foundation model and to the same tangent-filter Fisher object that governs the linear theory.

2 A Time-Series Foundation Model is an Autonomous World Model

2.1 The autonomous partially observed process

A univariate or multivariate time series is, in state-space form, the output of a partially observed dynamical system run *open loop*. There is a latent Markov state $X_t \in \mathsf{X} \subseteq \mathbb{R}^n$, an observation $O_t \in \mathsf{O} \subseteq \mathbb{R}^m$, and no action:

$$X_{t+1} \sim \mathcal{P}_\theta(\cdot \mid X_t), \qquad O_t \sim g_\theta(\cdot \mid X_t), \tag{2}$$

with unknown dynamics parameter $\theta \in \Theta \subseteq \mathbb{R}^p$. This is exactly the environment of the WAM/VLA model [1, §2] with the action A_t and the instruction ℓ removed. Write $H_t^- = (O_{1:t-1})$ for the history before O_t. Marginalizing the latent gives the *observation channel*

$$P_\theta(o_t \mid h_t^-) = \int_{\mathsf{X}} g_\theta(o_t \mid x_t)\, b_{t|t-1}^\theta(\mathrm{d}x_t), \qquad b_{t|t-1}^\theta = \mathbb{P}_\theta(X_t \in \cdot \mid H_t^-), \tag{3}$$

the predicted-belief mixture of the emission g_θ. A TSFM is a parameterized model of (3): given the past it emits a predictive law for the next observation.

Proposition 2.1 (The trajectory law has one factor)**.** *Under (2) the joint law of a length-T series factors as $\mathbb{P}_\theta(o_{1:T}) = \prod_{t=1}^T P_\theta(o_t \mid h_t^-)$, a single product of observation channels. There is no action factor: the WAM/VLA two-channel factorization [1, Prop. 2.1] degenerates to its first channel.*

Proof. Order the variables $o_1, o_2, \dots$ and apply the chain rule; the conditional law of O_t given $o_{1:t-1}$ is (3) by (2). The policy factor $\pi_\phi(a_t \mid \cdot)$ of [1] is absent because no A_t is generated. ∎

2.2 Chronos in the frame: tokenization is a quantized channel

CHRONOS [3] implements the channel (3) in three steps: (i) *mean-scale* the context, $\tilde{o} = o/s$ with $s = \frac{1}{C}\sum |o|$; (ii) *quantize* $\tilde{o}$ into one of B bins by $q : \mathbb{R} \to \{1, \dots, B\}$; (iii) model the resulting tokens with a categorical language model $p_\theta(z_{t+1} = i \mid z_{1:t})$ trained by cross-entropy. Steps (i)–(ii) compose the observation with a fixed, θ-free map $q \circ (\cdot/s)$; step (iii) is a discrete instance of (3). Thus CHRONOS is the observation channel *read through a quantizer*, and "regression via classification" is the categorical likelihood of that quantized channel. The next proposition is the Fisher consequence.

Proposition 2.2 (Tokenization is data processing on the FIM)**.** *Let $\mathcal{I}_{\theta\theta}$ be the Fisher information of the continuous observation channel (3) and $\mathcal{I}_{\theta\theta}^q$ that of its quantized (CHRONOS) counterpart $p_\theta(q(O_t) \mid h_t^-)$. Because q is a θ-free measurable map, $\mathcal{I}_{\theta\theta}^q \preceq \mathcal{I}_{\theta\theta}$ in the Loewner order, with the gap shrinking to zero as the bin width $\to 0$. Quantization can only lose information about the dynamics, and a direction unidentifiable in the continuous channel stays unidentifiable after tokenization.*

Proof. The score of the quantized channel is the conditional expectation of the continuous score given the quantized observation, $s_t^q = \mathbb{E}[s_t \mid q(O_t), h_t^-]$ (a θ-free coarsening); the Fisher data-processing inequality (the law of total covariance, as in [1, Cor. 3.4]) gives $\mathrm{Cov}(s_t^q) \preceq \mathrm{Cov}(s_t)$. As the partition refines, $q(O_t) \to O_t$ and the conditional expectation becomes the identity, so $\mathcal{I}_{\theta\theta}^q \uparrow \mathcal{I}_{\theta\theta}$. ∎

Proposition 2.2 says the *number of bins B* is an information knob: too coarse a quantizer throws away Fisher information about the very modes the model must learn. It also means the exact, continuous FIM $\mathcal{I}_{\theta\theta}$ is the right object to analyze—it is the ceiling that any tokenization approaches from below—which is what we do for the rest of the paper.

2.3 The Fisher information of a forecaster is the world block

Fix the true dynamics and consider identifying θ from the observation stream. Let $\mathcal{F}_t^- = \sigma(O_{1:t-1})$ and $s_t^\theta := \nabla_\theta \log P_\theta(O_t \mid H_t^-) \in \mathbb{R}^p$.

Proposition 2.3 (The single Fisher block)**.** *Under standard regularity the TSFM log-likelihood is $\ell_T(\theta) = \sum_t \log P_\theta(O_t \mid H_t^-)$; the score increments are martingale differences, $\mathbb{E}_\theta[s_t^\theta \mid \mathcal{F}_t^-] = 0$; and the Fisher information is*

$$\mathcal{I}_{\theta\theta}(\theta) = \sum_{t=1}^T \mathbb{E}_\theta\big[s_t^\theta (s_t^\theta)^\top\big] = \sum_{t=1}^T \mathbb{E}_\theta\big[\mathcal{I}_t^{\text{WAM}}(\theta)\big], \tag{4}$$

identical to the world block [1, Prop. 4.1] but now taken under the open-loop *law $\mathbb{P}_\theta$ (no policy occupancy). It is the* only *block: there is no policy score, hence no policy block and no cross block, and the occupancy coupling of [1, Prop. 6.3] is vacuous. Training a TSFM by next-observation prediction is maximum-likelihood system identification preconditioned by $\mathcal{I}_{\theta\theta}$.*

Proof. Proposition 2.1 makes the likelihood additive over observation factors; $\int P_\theta(o \mid h^-)\, \mathrm{d}o = 1$ gives $\int \nabla_\theta P_\theta = 0$ and the martingale property; vanishing cross-lag covariances (so the FIM is the diagonal sum) follow because s_s^θ is $\mathcal{F}_t^-$-measurable for $s < t$. There is no ϕ, so (1). The equivalence with maximum likelihood is $\nabla_\theta \mathcal{L} = \mathbb{E}[\sum_t s_t^\theta]$ and $-\mathbb{E}[\nabla_\theta^2 \mathcal{L}] = \mathcal{I}_{\theta\theta}$. ∎

The score is computed by the same *tangent filter* as the world block [1, §4.1]: propagating $(b_t^\theta, \nabla_\theta b_t^\theta)$ through the filtering recursion yields $\nabla_\theta P_\theta$ and hence s_t^θ; in the Gaussian/EKF case it is the differentiated Kalman filter, and for a learned recurrent or transformer latent it is backpropagation through the state update. A TSFM trainer computing gradients of the next-token loss is running this tangent filter whether it knows it or not.

3 The Context Embedding is an Approximate Information State

The TSFM's transformer summarizes the observed history into a context vector $z_t = \sigma_t(h_t)$ that it decodes into the next-token law. This is precisely an information-state generator [2]: z_t is (meant to be) sufficient to predict the next observation. In the exact linear–Gaussian case the sufficient statistic is the Kalman estimate; a finite transformer realizes only an (ε, δ)-AIS—approximately predictive to within ε in reward-free forecasting error and δ in its own update, measured in an integral probability metric (IPM). The next result is the forecasting instance of the AIS Fisher bound [1, Thm. 7.2].

Proposition 3.1 (AIS Fisher bound for a TSFM)**.** *Let $\hat{z}_t$ be an (ε, δ)-AIS whose predictive density is bounded below by $\underline{p} > 0$, with parameter-gradient approximated to within η_1 in the IPM sense and score bounded by $\bar{S}$. Then the FIM computed on the learned latent satisfies $\left\|\mathcal{I}_{\theta\theta} - \hat{\mathcal{I}_{\theta\theta}}\right\|_2 \leq 2T\bar{S}\,\beta$ with $\beta = \eta_1/\underline{p} + \bar{G}\,\rho_{\mathfrak{F}}(\cdot)\,\delta/\underline{p}^2$. A latent good enough for approximate forecasting is good enough to preserve identifiability, and—there being only one block—no spurious coupling can appear.*

Proof. The single-channel specialization of [1, Thm. 7.2]: apply the score-perturbation bound $\|s - \hat{s}\| \leq \eta_1/\underline{p} + \bar{G}(\hat{p} - p)/\underline{p}^2$ and $\|ss^\top - \hat{s}\hat{s}^\top\|_2 \leq 2\bar{S}\,\|s - \hat{s}\|$, then sum over t and substitute the IPM prediction bound. ■

Proposition 3.1 legitimizes analyzing the *exact* FIM $\mathcal{I}_{\theta\theta}$ as a proxy for the trained model's: the learned context perturbs the information by an IPM-controlled amount but cannot change its qualitative spectrum. We now compute that spectrum.

4 Stability Writes the Fisher Spectrum

We specialize to the linear–Gaussian (LQG) instance, where everything is closed form and every eigenvalue of the FIM belongs to a physical mode. The state is organized into K modes, each a damped oscillator; mode j has magnitude $r_j \in (0, 1)$ (its *stability*) and frequency ω_j, contributing a 2×2 block $r_j R(\omega_j)$ to A with R a rotation. Process noise $\Sigma_w = \sigma_w^2 I$ drives the state, sensor noise $\Sigma_v = \sigma_v^2 I$ corrupts the observation $O_t = CX_t + V_t$, and $C \in \mathbb{R}^{m\times n}$ reads out the modes with per-mode gain γ_j (so $\gamma_j = 0$ means mode j is unobserved). The dynamics parameter is the vector of modal magnitudes $\theta = (r_1, \ldots, r_K)$—the stabilities themselves.

4.1 The sensitivity Gramian

By Proposition 2.3 the FIM is the innovations Fisher information. Using the steady-state Kalman predictor with gain L, innovation covariance $\Sigma_\nu = CPC^\top + \Sigma_v$, and predictor closed loop $F = A(I - LC)$ (Schur stable), the per-step information about θ is the *parameter-sensitivity observability Gramian* of [1, Cor. 9.5],

$$\Gamma \;:=\; \mathcal{I}_{\theta\theta} \;=\; \mathbb{E}\big[(C\,\partial_\theta \hat{X}_{t|t-1})^\top \Sigma_\nu^{-1} (C\,\partial_\theta \hat{X}_{t|t-1})\big], \qquad \partial_\theta \hat{X}_{t+1|t} = F\,\partial_\theta \hat{X}_{t|t-1} + (\partial_\theta A)\hat{X}_{t|t}, \tag{5}$$

the tangent Kalman recursion of [1, App. A] specialized to the autonomous system. Equation (5) is exact and is what our numerics compute; its value is that the forcing $(\partial_{r_j} A)\hat{X}_{t|t}$ excites only mode j, so Γ is nearly diagonal in the modes and each of its eigenvectors points at one mode.

4.2 The stability–identifiability law

The heart of the paper is that the size of the information about mode j is set by that mode's stability margin.

Theorem 4.1 (Stability–identifiability law)**.** *For the modal LQG system the per-mode Fisher information obeys the AR(1) law*

$$\Gamma_{jj} \;\asymp\; \frac{\gamma_j^2\,\left\|c_j\right\|^2\,\sigma_w^2}{\sigma_v^2\,(1-r_j^2)}, \tag{6}$$

where $\left\|c_j\right\|^2$ *is the squared readout norm of mode* j*. Consequently the largest Fisher eigenvalue is governed by the slowest observed mode,*

$$\lambda_{\max}(\Gamma) \;\asymp\; \frac{1}{1-\rho(A)}, \tag{7}$$

and, since the fastest observed mode contributes an $O(1)$ *eigenvalue, the FIM condition number diverges as the stability margin vanishes,* $\kappa(\Gamma) = \lambda_{\max}/\lambda^{+}_{\min} \to \infty$ *as* $\rho(A) \to 1$*. Reducing the stability margin* increases *total information (*$\log\det\Gamma$, $\operatorname{tr}\Gamma$, $\lambda_{\max}$ *all grow) while* worsening *conditioning: learnability and conditioning are in tension.*

Proof sketch. A single mode in isolation is a two-dimensional AR(1): its stationary variance solves the scalar Lyapunov equation $\pi_j = r_j^2\pi_j + \sigma_w^2$, so $\pi_j = \sigma_w^2/(1-r_j^2)$. The one-step prediction $\hat{o} = c_j^\top \hat{x}_j$ has sensitivity $\partial_{r_j}\hat{x}_j$ proportional to the mode amplitude, so its stationary second moment scales with π_j; contracting with the innovation precision $\Sigma_v^{-1} \approx \sigma_v^{-2} I$ and the readout c_j gives (6) (the exact steady-state computation propagates $\partial_\theta \hat{X}$ through (5) and solves the resulting Lyapunov equation; the leading term is (6)). Taking the maximum over observed modes and using $\rho(A) = \max_j r_j$ gives (7); the fastest mode gives an $O(1)$ floor, so the ratio diverges. Monotonicity of log det, tr, and $\lambda_{\max}$ in the r_j follows from (6). Section 10 confirms (6)–(7) numerically to a log–log correlation of 0.97 and a $1/(1-\rho)$ fit. ∎

Equation (6) is the AR(1) folk fact $I(a) = 1/(1-a^2)$ [6] promoted to a foundation model: the slowest, most persistent modes near the unit circle carry almost all the Fisher information, and the fast modes near the origin carry almost none. This is why a forecaster trained on a nearly nonstationary series identifies its dominant dynamics so precisely—and why the same series' fast components are essentially unlearnable.

Remark 4.2 (Learnability versus control). The tension in Theorem 4.1 inverts the usual control intuition. A system near the unit circle is *hard to control* (the LQG cost inflates) but *easy to identify* (Fisher information inflates). A foundation model is a pure identifier, so it *benefits* from the regime a controller fears; the stability margin is a single dial trading the two.

5 Small Eigenvalues: Which Series a Foundation Model Cannot Learn

The eigenvectors of the small Fisher eigenvalues are the actionable output of the theory: they name the modes a forecaster cannot pin down, and—exactly as in [1, §11]—they say whether the cure is *more data* or a *new sensor*. The autonomous setting sharpens the dichotomy because a forecaster cannot explore: its only data knob is the length of the series.

Theorem 5.1 (Forecasting data–sensor dichotomy)**.** *Let* v *be a unit eigenvector of a small eigenvalue of* Γ*, pointing (by the modal structure of* (5)*) at a mode* j*. Exactly one of two cases holds.*

1. (excitation/data-limited) *Mode* j *is observed (*$\gamma_j > 0$*) but fast (*r_j *small), so* $\Gamma_{vv} = v^\top\Gamma v > 0$ *is small by* (6)*. The Cramér–Rao variance* $1/(T\,\Gamma_{vv}) \to 0$ *as the series length* $T \to \infty$*:* more data cures it*, at a rate set by* $1/(1-r_j^2)$*.*
2. (sensor-limited) *Mode* j *is unobserved (*$\gamma_j = 0$*), so* $v \in \ker\Gamma$ *and* $\Gamma_{vv} = 0$ *for* every *series length: no amount of data raises the eigenvalue. The direction is structurally unidentifiable and only a new observation channel that sees mode* j *(a row appended to* C*) cures it, by the additivity* $\Gamma^{C\cup c} = \Gamma^C + \Gamma_c$*.*

Moreover, whenever the observation map is fat*,* $m < n$ *(fewer sensors than latent states—the generic case for a real time series),* $\ker\Gamma \neq \{0\}$*: there is always a sensor-limited subspace, an* observability ceiling *on what univariate/low-dimensional forecasting can learn.*

Proof. The modal structure of (5) gives $\Gamma_{vv} = \Gamma_{jj}$ up to the eigenvector's concentration; (6) gives case 1 and the rate. For case 2, $\gamma_j = 0$ makes mode j's readout $c_j = 0$, so $C\partial_{r_j}\hat{X} \equiv 0$ and the integrand of (5) vanishes identically; hence $\Gamma_{jj} = 0$ irrespective of T, and additivity of Fisher information over conditionally independent channels gives the sensor cure. When $m < n$, C has a nontrivial kernel; modes lying in it have $c_j = 0$ and populate $\ker\Gamma$. ∎

Theorem 5.1 is the forecasting reading of the WAM/VLA dichotomy [1, Thm. 11.3]: a small Fisher eigenvalue on a foundation model is a message. If its mode is observed, collect a longer series (or, in the multi-series regime, more series that excite that mode); if its mode is unobserved, no corpus will ever teach the model to forecast it—acquire a new measurement channel. The fat-C corollary is the precise sense in which a univariate forecaster faces an observability ceiling: it can only learn the part of the dynamics its single channel sees.

Example 5.2 (Position without velocity)**.** A second-order mechanical mode observed only in position ($C = [1, 0]$) has its velocity coordinate unobserved; the corresponding parameter direction is sensor-limited, and no length of position-only data identifies it, whereas a velocity channel lifts it immediately. Section 10 realizes this at scale: eight modes are left unobserved and produce exactly eight null Fisher directions, cured only by adding channels that see them.

6 Learnability: From the Fisher Spectrum to Learning Curves

The FIM spectrum predicts not just *whether* but *how fast* a foundation model learns each part of the dynamics, through the Cramér–Rao bound and the martingale central limit theorem for the next-token MLE.

Proposition 6.1 (Learning curve and per-mode risk)**.** *Let $\hat{\theta}_T$ be the next-token maximum-likelihood estimate from a series of length T. Under identifiability of the observed modes, $\sqrt{T}(\hat{\theta}_T - \theta^\star) \Rightarrow \mathcal{N}(0, \bar{\Gamma}^{-1})$ on the observed subspace, where $\bar{\Gamma}$ is the per-step FIM. Consequently:*

1. *the estimation variance along mode j is $1/(T\bar{\Gamma}_{jj}) \asymp \sigma_v^2(1 - r_j^2)/(T\gamma_j^2 \left\|c_j\right\|^2)$:* slow modes are learned first, fast modes last, unobserved modes never*;*
2. *the excess h-step forecast error along mode j is bounded by $S_{j,h}^2/(T\bar{\Gamma}_{jj})$, with $S_{j,h} = \partial_{r_j}(h\text{-step predictor})$ the prediction sensitivity, so early in training aggregate skill is dominated by the slow, high-Fisher modes and rises with* $\log\det\bar{\Gamma}$*;*
3. *the sloppy directions ($\lambda_j \downarrow$) require $T \gtrsim 1/(\varrho^2\lambda_j)$ samples to reach relative precision ϱ, so the number of learnable modes grows only logarithmically in T given the geometric Fisher spectrum of* (6)*.*

Proof. The joint score is a martingale (Proposition 2.3) with predictable quadratic variation $T\bar{\Gamma}$; the martingale CLT and a Taylor expansion of the stationarity condition give the normal limit with covariance $\bar{\Gamma}^{-1}$ (restricted to the identifiable subspace, $\ker\bar{\Gamma}$ excluded). Claim 1 is the diagonal of $\bar{\Gamma}^{-1}$ with (6); claim 2 is the delta method on the h-step predictor; claim 3 is the CRB threshold $v^\top\bar{\Gamma}^{-1}v \geq 1/\lambda_j$. ∎

Proposition 6.1 converts the sloppy Fisher spectrum into a concrete curriculum: a foundation model learns a system's modes in *decreasing order of stability*, each at a rate proportional to its Fisher eigenvalue, and reaches an accuracy floor set by the sloppiest modes it can still see. Section 10 shows a trained CHRONOS-lite exhibiting exactly this: forecast skill rising with data and with $\log\det\Gamma$, and a stiff/sloppy learnability ratio equal to the Fisher eigenvalue ratio.

7 When the Second Block Returns: Covariates and Control

The collapse (1) is exact for *autonomous* forecasting. Real forecasting problems often include *covariates*—exogenous inputs u_t (promotions, weather, control actions) that drive the series. A covariate is an action the forecaster does not choose but must condition on, and it re-activates the second channel.

Proposition 7.1 (Covariate forecasting re-opens the policy block)**.** *Suppose the state obeys $X_{t+1} \sim \mathcal{P}_\theta(\cdot \mid X_t, U_t)$ with exogenous $U_t \sim \pi_\phi(\cdot \mid H_t)$ generated by a (possibly unknown) mechanism ϕ. Then the trajectory law regains the two-channel factorization of [1, Prop. 2.1]: an observation channel $P_\theta(o_t \mid h_t^-, u_{t-1})$ and a covariate channel $\pi_\phi(u_t \mid h_t)$. If ϕ is estimated, the joint FIM regains its block structure* $\mathrm{blkdiag}(\mathcal{I}_{\theta\theta}, \mathcal{I}_{\phi\phi})$ *and, crucially, its* occupancy coupling *[1, Prop. 6.3]: the covariate distribution sizes $\mathcal{I}_{\theta\theta}$ (which dynamics the data excite) and the dynamics size $\mathcal{I}_{\phi\phi}$. A TSFM with covariates is therefore back in the full two-block theory; pure forecasting is the ϕ-free corner.*

Proof. Re-instate the action factor in Proposition 2.1; the interleaved-filtration argument of [1, Thm. 6.1] gives block-diagonality and [1, Prop. 6.3] the occupancy coupling. Deleting U_t recovers Proposition 2.3. ∎

Thus the WAM/VLA/TSFM trichotomy is a single axis: no exogenous input (pure TSFM, one block $\mathcal{I}_{\theta\theta}$); exogenous but not chosen (covariate TSFM, two blocks, coupled); chosen by a learned policy under instruction (VLA, two blocks, and the co-design of [1, §6]). Stability and observability govern $\mathcal{I}_{\theta\theta}$ in all three; what the covariate/control adds is the occupancy that *sizes* $\mathcal{I}_{\theta\theta}$, and its own identifiability $\mathcal{I}_{\phi\phi}$.

8 The Model Zoo: Transition Structures and Hyperparameters

The theory of the previous sections, and the chaotic theory of the next, are anchored to a concrete corpus. We generate time-series data from five environments chosen to *sweep the stability-to-chaos axis*: a stable linear Gaussian process and four nonlinear oscillator/field systems whose top Lyapunov exponents (measured in Section 9) run from contracting through neutral to strongly positive. Each environment is the time-Δ sampling of a (stochastic or deterministic) dynamical system on $\mathbb{R}^N$; a TSFM sees only the observation series $o_t = g(x_t) + \text{(noise)}$. Table 2 collects the transition structures, Table 3 every hyperparameter, and Table 4 the CHRONOS-lite recipe. All values are those in the accompanying code (`environment/` and `tsfm/`).

8.1 Generative environments

(E1) Modal LQG—linear–Gaussian, stable. A partially observed linear system in real modal form,

$$X_{t+1} = AX_t + W_t, \qquad O_t = CX_t + V_t, \qquad W_t \sim \mathcal{N}(0, \sigma_w^2 I_n),\ V_t \sim \mathcal{N}(0, \sigma_v^2 I_m), \tag{8}$$

with $A = \bigoplus_{j=1}^{K} r_j R(\omega_j)$ block-diagonal in K damped oscillatory modes, $R(\omega) = \left(\begin{smallmatrix} \cos\omega & \sin\omega \\ -\sin\omega & \cos\omega \end{smallmatrix}\right)$, so $n = 2K$. Modal magnitudes r_j are log-spaced in $[r_{\text{lo}}, \rho]$; the readout C has per-mode gain tilted toward the slow modes by $(r_j/\rho)^\beta$, and the n_{h} fastest modes are left *unobserved* ($c_j = 0$). Every Lyapunov exponent is $\log r_j < 0$, so the system is globally contracting; this is the instance analyzed exactly in Sections 2 to 6 and Section 10.

(E2) Multi-population Kuramoto—coupled phase oscillators. N phases on the circle,

$$\dot\theta_i = \omega_i + \sum_{j=1}^{N} K_{ij} \sin(\theta_j - \theta_i), \qquad o_t = \theta_t \bmod 2\pi, \tag{9}$$

with a block coupling $K_{ij} = \kappa_{\text{intra}}/N$ inside each of P communities and κ_{inter}/N between them, and natural frequencies $\omega_i \sim \mathcal{N}(0, \varsigma^2)$. Strong intra-community coupling drives synchronization into P clusters, collapsing the macroscopic dynamics onto a $\approx P$-dimensional manifold.

(E3) Cluster Stuart–Landau—coupled Hopf oscillators. N complex amplitudes near a Hopf bifurcation,

$$\dot z_i = (\lambda_i + i\omega_i) z_i - |z_i|^2 z_i + \sum_{j=1}^{N} K_{ij}(z_j - z_i), \qquad o_t = \Re\, z_t, \tag{10}$$

$z_i \in \mathbb{C}$, with the same block coupling as (E2) over P clusters, $\lambda_i \equiv \lambda > 0$ and $\omega_i \sim \mathcal{N}(\bar\omega, \varsigma^2)$. Diffusive coupling synchronizes each cluster onto a common limit cycle, giving a $\approx P$-dimensional (quasi-)periodic attractor.

(E4) Kuramoto–Sivashinsky—spatiotemporal chaos. The scalar field $u(x,t)$ on $[0, L)$ with periodic boundaries,

$$\partial_t u + u\,\partial_x u + \partial_x^2 u + \partial_x^4 u = 0, \tag{11}$$

discretized by a Fourier–Galerkin pseudo-spectral method on N modes: with $\hat u_k$ the Fourier coefficients and k the wavenumbers, $\partial_t \hat u_k = (k^2 - k^4)\hat u_k - \frac{ik}{2}\widehat{u^2}_k$, advanced by the semi-implicit (IMEX) Euler step $\hat u_k^+ = (1 - \Delta(k^2 - k^4))^{-1}\big(\hat u_k - \frac{ik}{2}\Delta\,\widehat{u^2}_k\big)$; the implicit treatment of the stiff k^4 hyper-diffusion is what allows the coarse Δ. The observation is the field on the grid, $o_t = u(\cdot, t)$. At $L = 22$ the attractor is low-dimensional but chaotic.

Table 2: Transition structures of the five generative environments. ODE systems are advanced by classical RK4, the stiff PDE by IMEX Euler; the observation cadence is $\Delta_s = \Delta \cdot S_{save}$.

System	Transition / vector field	State	Regime (by λ_1)
LQG (E1)	$X_{t+1} = AX_t + W_t,\ O_t = CX_t + V_t,$ $A = \bigoplus_j r_j R(\omega_j)$	$\mathbb{R}^n,\ n = 2K$	contracting, $\lambda_1 < 0$
Kuramoto (E2)	$\dot{\theta}_i = \omega_i + \sum_j K_{ij} \sin(\theta_j - \theta_i)$	$\mathbb{T}^N$	synchronized, $\lambda_1 \approx 0^+$
Stuart–Landau (E3)	$\dot{z}_i = (\lambda + i\omega_i)z_i - \lvert z_i\rvert^2 z_i + \sum_j K_{ij}(z_j - z_i)$	$\mathbb{C}^N$	limit cycle, $\lambda_1 \approx 0$
KS (E4)	$\partial_t u = -u\partial_x u - \partial_x^2 u - \partial_x^4 u$	$\mathbb{R}^N$ (grid)	low-dim chaos
Lorenz–96 (E5)	$\dot{y}_i = (y_{i+1} - y_{i-2})y_{i-1} - y_i + F$	$\mathbb{R}^N$	high-dim chaos

Table 3: Environment hyperparameters (as configured in `environment/`). Δ is the integrator step, S_{save} the sub-sampling stride, $\Delta_s = \Delta S_{save}$ the observation spacing. The corpus uses $N_{tr} = 100$ training and $M_{te} = 10$ test trajectories of length $T = 200$ per system.

System	Dimensions & couplings	Integration
LQG (E1)	K=100 modes, n=200, m=100; ρ=0.90, r_{lo}=0.15; σ_w=1.0, σ_v=0.1; n_h=8 hidden; gain $\in [0.4, 1.0]$, β=2.5	Δ=1, S_{save}=1
Kuramoto (E2)	N=200, P=5; κ_{intra}=5.0, κ_{inter}=0.1; $\omega \sim \mathcal{N}(0, 0.5^2)$	Δ=0.05, S_{save}=2
Stuart–Landau (E3)	N=200, P=7; κ_{intra}=2.0, κ_{inter}=0.05; λ=1, $\omega \sim \mathcal{N}(2, 0.2^2)$	Δ=0.05, S_{save}=2
KS (E4)	N=200 modes, domain L=22.0	Δ=10^{-3}, S_{save}=50
Lorenz–96 (E5)	N=40, forcing F=8.0	Δ=0.01, S_{save}=5

(E5) Lorenz–96—high-dimensional chaos. N sites on a ring with advective coupling and constant forcing,

$$\dot{y}_i = (y_{i+1} - y_{i-2})\, y_{i-1} - y_i + F, \qquad i \in \mathbb{Z}/N\mathbb{Z}, \qquad o_t = y_t, \tag{12}$$

at the canonical chaotic forcing $F = 8$. Unlike (E2)–(E4) this system is *not* designed to be low-dimensional: at $N = 40, F = 8$ it has many positive Lyapunov exponents (Section 9), making it the stress test for the observability ceiling of Theorem 9.6.

8.2 Tokenizer, backbone, and objective (chronos-lite)

A single *univariate* foundation model is trained across all channels of all systems (the global-model regime). It follows the chronos recipe of Section 2 in three steps. **(i) Mean scaling and quantization.** For a context window $\mathbf{c}$, compute $s = \frac{1}{|\mathbf{c}|}\sum |c|$, scale $\tilde{y} = y/s$, and quantize into one of B uniform bins over $[q_{lo}, q_{hi}]$ in scaled space, $q_t = \text{Quantize}(y_t/s; B) \in \{1, \dots, B\}$ (the fixed, θ-free channel of Proposition 2.2). **(ii) Windowing.** Slice each tokenized channel into length-C contexts with stride S, paired with the next token. **(iii) Language-model objective.** A decoder-only Transformer P_θ over the B-symbol vocabulary is trained by next-token cross-entropy,

$$\mathcal{L}(\theta) = -\mathbb{E}_{(\mathbf{c}_t, q_{t+1})\sim\mathcal{D}}\big[\log P_\theta(q_{t+1} \mid \mathbf{c}_t)\big], \qquad P_\theta(\cdot \mid \mathbf{c}_t) = \text{Softmax}\big(\text{head}(\mathbf{h}_C)\big), \tag{13}$$

with $\mathbf{h}_C$ the final causal hidden state—the realized approximate information state (Section 3, and Theorem 9.2 below). Forecasts are autoregressive token samples, de-quantized and un-scaled, or the categorical expected value (the minimum-MSE point forecast). Table 4 lists the architecture and optimizer settings; the exact-FIM study of Section 10 uses the smaller per-system variant noted there.

Table 4: The CHRONOS-lite tokenizer, backbone, and training recipe (`tsfm/chronos_lite.py`, `tsfm/config.json`). The cross-system foundation model is the main configuration; the LQG-only Fisher study of Section 10 uses the compact variant.

Component	Cross-system model	LQG Fisher study
Tokenizer	mean-scale + uniform quantize	(same)
Vocabulary B	256 bins	128 bins
Scaled range $[q_{\text{lo}}, q_{\text{hi}}]$	$[-6, 6]$	$[-6, 6]$
Context length C	32	40
Window stride S	8	8
Backbone	causal Transformer (decoder-only)	(same)
Embedding d_{model}	256	64
Layers / heads	2 / 8	2 / 8
Feed-forward / act.	$4d_{\text{model}}$ / GELU	$4d_{\text{model}}$ / GELU
Dropout, norm	0.1, LayerNorm	(same)
Optimizer	AdamW, lr $3{\times}10^{-3}$, wd 10^{-4}	(same)
Grad clip, batch, epochs	1.0, 32, 30	1.0, 256, ~6
Forecast	AR sampling (30 paths, τ=1) / E-value	(same)

9 Chaotic Dynamics, the Information State, and Foundation-Model Predictability

The exact spectral theory of Sections 4 to 6 is linear–Gaussian: it identifies a *parameter* θ and its Fisher matrix is written by stability margins $1 - r_j^2$. Four of the five environments of Section 8 are *nonlinear*, and (E4)–(E5) are *chaotic*. This section develops the mathematics a foundation model faces on such systems and threads it through the same approximate-information-state (AIS) latent. Three facts change relative to the linear case, and each is a theorem below: the sufficient statistic is a low-dimensional *delay embedding* of the attractor rather than a Kalman mean (Theorem 9.2); forecast skill is bounded not by a sloppy spectrum but by a hard *Lyapunov horizon* (Theorem 9.4); and the observability null space becomes a *dynamically diverging* subspace (Theorem 9.6). A fourth result ties tokenization to the metric entropy (Proposition 9.8), and a fifth shows the linear and chaotic theories are one axis in the top Lyapunov exponent (Proposition 9.10).

9.1 The reward-free approximate information state

We first state, for self-containedness, the forecasting specialization of the AIS of Subramanian et al. [2]. Let $\mathcal{H}_t = \sigma(O_{1:t})$ and let $d_{\mathfrak{F}}$ be an integral probability metric generated by a function class $\mathfrak{F}$ (Wasserstein-1 when $\mathfrak{F}$ is 1-Lipschitz).

Definition 9.1 ((ε, δ)-AIS for forecasting)**.** A statistic $z_t = \sigma_t(O_{1:t}) \in \mathsf{Z}$ with predictive kernel $\hat{P}(\cdot \mid z)$ on O and update $z_t = \psi(z_{t-1}, O_t)$ is an (ε, δ)-*approximate information state* if for all t: (AP1) predictive sufficiency, $d_{\mathfrak{F}}\big(\mathrm{Law}(O_{t+1} \mid \mathcal{H}_t),\ \hat{P}(\cdot \mid z_t)\big) \le \varepsilon$; and (AP2) recursive sufficiency, $d_{\mathfrak{F}}\big(\mathrm{Law}(z_{t+1} \mid \mathcal{H}_t),\ \psi(z_t, \cdot)_\# \hat{P}(\cdot \mid z_t)\big) \le \delta$. It is *exact* when $\varepsilon = \delta = 0$, i.e. z_t is a predictive sufficient statistic for the future.

This is the reward-free reading of the AIS: the "reward" a forecaster must predict is the next observation itself, so predictive sufficiency for O_{t+1} replaces sufficiency for a reward. The TSFM's context hidden state $\mathbf{h}_C$ of (13) is a learned candidate z_t; the theorems below say what the *ideal* z_t is for a chaotic system and how large it must be.

9.2 Standing assumptions and the Lyapunov data of the corpus

Assumption 1 (Chaotic regime)**.** The observation series is the time-Δ sampling of a dissipative system: a $C^{1+\alpha}$ diffeomorphism $f : \mathcal{M} \to \mathcal{M}$ (the flow map on a finite-dimensional state space $\mathcal{M} \subseteq \mathbb{R}^n$; for the PDE, its Fourier–Galerkin truncation) with a compact attractor $\mathcal{A}$ carrying an ergodic SRB measure μ; observations $O_t = g(X_t) + V_t$ with $g \in C^1$ and $V_t \sim \mathcal{N}(0, \sigma_v^2 I_m)$ i.i.d. By Oseledets' multiplicative ergodic theorem [16] the Lyapunov exponents

Table 5: Measured Lyapunov data of the corpus (per unit model-time). LQG is linear, so $\lambda_1 = \log \rho$ exactly; Lorenz–96 is measured by full-spectrum Benettin QR (hence n_+, D_{KY}, h_{KS}); the nonlinear oscillators/field by the leading-exponent method. Attractor dimensions marked † are the design targets of Section 8 (box/Lyapunov dimension); D_{KY} for Lorenz–96 is measured. The top exponent sweeps from contracting through neutral to strongly chaotic.

System	λ_1 (time^{-1})	n_+	D_{KY}	h_{KS} (nats/time)	Regime
LQG (E1)	−0.105	0	—	0	contracting (fully forecastable)
Stuart–Landau (E3)	$\approx +0.010$	$\lesssim 1$	$\approx 7^{\dagger}$	≈ 0	neutral / limit cycle
Kuramoto (E2)	$\approx +0.033$	few	$\approx 5^{\dagger}$	small	edge / weak chaos
KS (E4)	$\approx +0.030$	few	5–8†	small	low-dim chaos ($T_\lambda \approx 34$)
Lorenz–96 (E5)	+1.68	13	27.1	10.5	high-dim chaos ($T_\lambda \approx 0.59$)

$\lambda_1 \geq \cdots \geq \lambda_n$ and the measurable splitting $T_x\mathcal{M} = E^u(x) \oplus E^c(x) \oplus E^s(x)$ (over $\lambda_i > 0, = 0, < 0$) exist μ-a.e.; write $n_+ = \dim E^u$, $n_0 = \dim E^c$. The system is *chaotic* iff $\lambda_1 > 0$. Define the Kolmogorov–Sinai entropy $h_{\mathrm{KS}}(\mu)$ and the Kaplan–Yorke dimension $D_{\mathrm{KY}} = \kappa + |\lambda_{\kappa+1}|^{-1} \sum_{i \leq \kappa} \lambda_i$ with $\kappa = \max\{k : \sum_{i \leq k} \lambda_i \geq 0\}$.

To make the constants below concrete we measured the Lyapunov spectra of the corpus directly from the repository's integrators (Benettin QR with a finite-difference tangent map for the full Lorenz–96 spectrum; the two-trajectory method for the leading exponent of the others). Table 5 reports them. The five systems realize a clean monotone sweep of λ_1 across zero, which Proposition 9.10 turns into the organizing axis of the paper.

9.3 The context window is a delay-embedding information state

The first structural fact is that, for a chaotic attractor, the ideal AIS is not a Kalman mean but a *delay embedding*, and its size is the attractor's dimension—not the ambient n. This is why a modest-context foundation model can forecast a nominally 200-dimensional field: it only has to embed a 5–10 dimensional attractor.

Theorem 9.2 (The context is a delay-embedding AIS). *Under Assumption 1, let $d_B = \dim_B \mathcal{A}$ be the box-counting dimension of the attractor and fix an integer delay window $w \geq 2d_B + 1$. Let $z_t = (O_t, O_{t-1}, \ldots, O_{t-w+1}) \in \mathbb{R}^{mw}$ be the length-w observation context and $\Phi_w : \mathcal{A} \to \mathbb{R}^{mw}$, $x \mapsto (g(x), g(f^{-1}x), \ldots, g(f^{-(w-1)}x))$, the delay map.*

1. (exact, noise-free) *For generic (f, g) in the sense of Sauer–Yorke–Casdagli, Φ_w is a bi-Lipschitz embedding of $\mathcal{A}$; hence in the limit $\sigma_v \to 0$ the state is recovered, $x_t = \Phi_w^{-1}(z_t)$, and z_t is an* exact *information state (Definition 9.1 with $\varepsilon = \delta = 0$): the one-step predictor $\hat{O}_{t+1} = g(f(\Phi_w^{-1} z_t))$ equals the Bayes predictor.*
2. (approximate, noisy) *If $F := \Phi_w^{-1}$ is* $\mathrm{Lip}(F)$*-Lipschitz on a neighborhood of $\Phi_w(\mathcal{A})$, then z_t is an (ε, δ)-AIS in Wasserstein-1 with $\varepsilon, \delta \leq L_\star\, \sigma_v \sqrt{m}$, where $L_\star = c\ \mathrm{Lip}(g \circ f)\ \mathrm{Lip}(F)$ for a universal constant c.*
3. (dimension) *Predictive sufficiency needs only $mw = O(m\, d_B)$ context coordinates, and the intrinsic AIS dimension is the attractor dimension d_B (estimated by D_{KY}),* independent of the ambient dimension *n.*

Proof. (1) The fractal Takens theorem of Sauer–Yorke–Casdagli [15] (extending Takens [14]) states that for $w > 2d_B$ a generic delay-coordinate map is injective and immersive on a compact invariant set; injectivity of a continuous map on a compact set gives a homeomorphism onto its image, and smoothness of f, g upgrades this to bi-Lipschitz on $\mathcal{A}$. Hence $F = \Phi_w^{-1}$ exists and recovers x_t; since (X_t) is Markov, the future law depends on the history only through x_t, so a statistic determining x_t is predictive-sufficient and $\hat{O}_{t+1} = g(f(x_t))$ is the Bayes one-step predictor. (2) With noise, $z_t = \Phi_w(x_t) + e_t$ where e_t stacks w observation noises, $\mathbb{E}\|e_t\| \leq \sigma_v \sqrt{mw}$ (absorb $\sqrt{w}$ into c). The map $z \mapsto g(f(F(z)))$ is Lipschitz with constant $\mathrm{Lip}(g \circ f)\,\mathrm{Lip}(F)$, so the predictive-mean perturbation is at most $L_\star \sigma_v \sqrt{m}$; because Wasserstein-1 is dominated by the mean transport $\mathcal{W}_1(\mathrm{Law}(a), \mathrm{Law}(a')) \leq \mathbb{E}\|a - a'\|$, both (AP1) and (AP2) (the latter using the same Lipschitz update ψ) hold with that bound. (3) is the coordinate count mw with $w = \lceil 2d_B + 1 \rceil$, and the observation that $F(z_t) \in \mathcal{A}$ lives on a set of dimension d_B. ■

Remark 9.3 (Why context beats width). Theorem 9.2 explains a robust empirical fact about TSFMs on physical data: a short context can be sufficient when the attractor is low-dimensional (the oscillator systems, $d_B \approx 5$–7), whereas a system with a large attractor (Lorenz–96, $D_{\mathrm{KY}} \approx 27$) needs $w \gtrsim 55$ delays for a univariate sensor—longer than the $C = 32$ context of Table 4, so that model is *structurally* under-contextualized on (E5), independent of training.

9.4 The Lyapunov predictability horizon

The second fact is a hard ceiling on forecast skill. In the linear case sloppy modes are learned *slowly*; in the chaotic case the future is *unknowable* past a horizon set by λ_1, for every model.

Theorem 9.4 (Lyapunov horizon; forecast-risk floor)**.** *Under Assumption 1 with $\lambda_1 > 0$, suppose the analysis-consistent set $\mathcal{U}_t = \{x : x$ compatible with $\mathcal{H}_t$ to within the noise$\}$ contains a segment of length $\geq \delta_0$ along the top Oseledets direction $u_1(x_t)$ (generically true with $\delta_0 \asymp \mathrm{Lip}(F)\sigma_v$ by Theorem 9.2), and let g have non-degenerate differential $c_g > 0$ along the forward image of u_1. Then for every $\eta > 0$, all large h, and every $\mathcal{H}_t$-measurable h-step predictor $\hat{O}_{t+h}$,*

$$\sup_{x \in \mathcal{U}_t} \left\| g(f^h x) - \hat{O}_{t+h} \right\|^2 \;\geq\; \tfrac{1}{4}\, c_g^2 \, \min\!\Big(\delta_0^2\, e^{2(\lambda_1 - \eta)h},\; \mathrm{diam}\, g(\mathcal{A})^2\Big). \tag{14}$$

Consequently positive skill $R_h^2 = 1 - \mathrm{MSE}(h)/\sigma_{\mathcal{A}}^2 > 0$ is attainable only within the Lyapunov (predictability) horizon

$$h \;\lesssim\; \tau_\star = \frac{1}{\lambda_1} \log \frac{\mathrm{diam}\, g(\mathcal{A})}{\delta_0} \;\asymp\; \frac{1}{\lambda_1} \log \frac{1}{\sigma_v}. \tag{15}$$

The horizon depends only on the dynamics and the sensor: it is invariant to architecture, to context length beyond w, to corpus size, and to tokenizer resolution, and grows only logarithmically in $1/\sigma_v$.

Proof. Take $x, x' \in \mathcal{U}_t$ separated by δ_0 along $u_1(x_t)$. By Oseledets [16], $\frac{1}{h} \log \|Df^h(x_t)u_1\| \to \lambda_1$ μ-a.e., so $\|Df^h(x_t)u_1\| \geq e^{(\lambda_1 - \eta)h}$ for $h \geq h_0(\eta, x_t)$; while the perturbation stays in the linear regime, $\|f^h x - f^h x'\| \geq \delta_0 e^{(\lambda_1-\eta)h}$, saturating at $O(\mathrm{diam}\,\mathcal{A})$ once it exits. Applying g (differential c_g along the image) gives $\|g(f^h x) - g(f^h x')\| \geq c_g \min(\delta_0 e^{(\lambda_1-\eta)h}, \mathrm{diam}\, g(\mathcal{A}))$. Any single value $\hat{O}_{t+h}$ is at distance $\geq \frac{1}{2}$ this separation from one of $g(f^h x), g(f^h x')$ (triangle inequality), giving (14). Finally $R_h^2 > 0$ requires $\mathrm{MSE}(h) < \sigma_{\mathcal{A}}^2 = \Theta(\mathrm{diam}\, g(\mathcal{A})^2)$, which fails once the min in (14) saturates, i.e. for $h > \tau_\star$ in (15). ∎

Remark 9.5 (The floor is the classic error-growth curve; Bayes version). Inequality (14) is the minimax form of the Lorenz error-growth law [21]: exponential separation at rate λ_1 until saturation at the climatological variance. The same bound holds for the Bayes risk whenever the analysis posterior places $\Omega(1)$ mass near both endpoints of the δ_0-segment. Numerically, $\tau_\star$ is ≈ 0.59 time for Lorenz–96 but ≈ 34 time for KS (Table 5): the same foundation model is useful many steps ahead on the smooth field and only a step or two ahead on the stiff one, exactly in inverse proportion to λ_1.

9.5 The dynamic observability ceiling

In the linear theory (Theorem 5.1) an unobserved mode gives a *static* null direction of the parameter FIM. For a chaotic system the analogue is sharper: an unobserved *unstable* direction makes the state error diverge exponentially. The relevant object is the state (data-assimilation) Fisher information—the observability Gramian of the tangent filter—rather than the parameter FIM, but it is computed by the *same* tangent recursion of (5), now differentiated with respect to the state rather than θ.

Theorem 9.6 (Unstable-subspace observability; AIS tracks iff it sees E^u)**.** *Consider the tangent estimation problem along a μ-typical orbit, with tangent map $M_t = Df(X_t)$ and observation Jacobian $C_t = Dg(X_t)$, and let $E^u \oplus E^c$ be the non-negative Oseledets subspace* ($\dim = n_+ + n_0$).

1. (bounded tracking) *If $E^u \oplus E^c$ is uniformly completely observable through $\{(C_t, M_t)\}$, the information filter has a unique bounded stationary error covariance supported on $E^u \oplus E^c$; the delay-AIS of Theorem 9.2 tracks the state with error uniformly bounded in t, despite $\lambda_1 > 0$. The stable subspace E^s is self-correcting: its error decays at the stable rates regardless of observation.*
2. (divergence) *If some unstable direction $u \in E^u$ with exponent $\lambda_u > 0$ is uniformly unobservable (C_t annihilates the transported u along the orbit), then every filter/AIS has error along u obeying* $\liminf_t \frac{1}{t} \log \mathbb{E}\|e_t \cdot u\| \geq \lambda_u > 0$*: it diverges exponentially, and no context length or corpus removes it.*
3. (fat-sensor chaotic ceiling) *Bounded tracking requires the observations to span the $n_+ + n_0$ non-negative directions; hence whenever $m < n_+$—automatic for a univariate or low-m forecaster of a high-dimensional chaos such as Lorenz–96, where $n_+ = 13$—some unstable direction is unobserved and (2) applies.*

Proof. This is the "assimilation in the unstable subspace" phenomenon [23, 24, 25]. For the linear tangent cocycle the Kalman information matrix asymptotically annihilates the strictly stable directions (their errors contract under M_t irrespective of data), so the non-trivial filtering problem lives on $E^u \oplus E^c$; uniform observability of that subspace yields a projected Riccati map that is a uniform contraction on the cone of positive-definite operators, giving a unique bounded fixed point (claim 1). For claim 2, an unobserved direction receives zero innovation gain, so its error obeys the homogeneous recursion $e_{t+1} = M_t e_t$ restricted to u, whose growth rate is λ_u by Oseledets. Claim 3 is the rank count $\dim(E^u \oplus E^c) = n_+ + n_0 > m$ when $m < n_+$. ■

Remark 9.7 (Static null vs. dynamic divergence). Theorems 5.1 and 9.6 are the two faces of one observability ceiling. When $\lambda_1 < 0$ (LQG) an unobserved mode is a *static* kernel direction of the parameter FIM—unidentifiable but harmless to forecasting a stable series. When $\lambda_1 > 0$ an unobserved *unstable* mode is a *dynamic* defect: its estimation error grows like $e^{\lambda_u t}$, so it destroys state tracking, not merely parameter identification. Chaos converts the sensor-limited null space from a nuisance into a hard wall.

9.6 Metric entropy sets the tokenizer information rate

Tokenization was priced in Proposition 2.2 as data processing on the parameter FIM. For a chaotic system it acquires a second, sharper role: the vocabulary must be large enough to convey the information the dynamics *creates* each step.

Proposition 9.8 (Pesin bound on the tokenizer). *Under Assumption 1, Pesin's formula for the SRB measure gives $h_{\mathrm{KS}}(\mu) = \sum_{i:\lambda_i>0} \lambda_i$ [17, 18, 19]. To hold the filtering/AIS error stationary at fixed precision, the observation channel must convey mutual information about the state at rate $\geq h_{\mathrm{KS}}(\mu)$ per unit time. A* CHRONOS *tokenizer emitting m tokens per observation step from a B-symbol vocabulary has per-step capacity $\leq m \log B$; hence a necessary condition for the tokenized AIS to track the state is*

$$m \log B \;\geq\; h_{\mathrm{KS}}(\mu)\,\Delta_{\mathrm{s}} \;=\; \Big(\sum_{\lambda_i>0} \lambda_i\Big)\,\Delta_{\mathrm{s}} \qquad (\textit{per sampled step, spacing } \Delta_{\mathrm{s}}). \tag{16}$$

When B is too small the quantized channel is throttled below the entropy production and forecast skill collapses even within *the Lyapunov horizon; coarser sampling (larger Δ_{s}) tightens the bound. Thus B is lower-bounded by the system's entropy rate, not merely by a target numerical resolution.*

Proof. By the Kolmogorov–Sinai/Pesin identity the dynamics generates h_{KS} nats of new state uncertainty per unit time in the unstable subspace; a stationary posterior entropy requires the observation to remove at least as much, i.e. $I(O_{t+1}; X_{t+1} \mid \mathcal{H}_t) \geq h_{\mathrm{KS}}\Delta_{\mathrm{s}}$ per sampled step. The tokenized observation is a vector of m symbols from an alphabet of size B, whose entropy—hence the mutual information it can carry—is at most $m \log B$. Combining gives (16). ■

Remark 9.9 (The corpus is comfortably above the Pesin floor). For Lorenz–96, $h_{\mathrm{KS}}\Delta_{\mathrm{s}} \approx 10.5 \times 0.05 \approx 0.53$ nats/step, far below the full-state token budget $m \log B = 40 \log 256 \approx 222$ nats; the tokenizer is not the bottleneck—observability is (Theorem 9.6). A *univariate* sensor ($m = 1$) still clears the rate bound ($\log 256 \approx 5.5$ nats) yet fails on (E5) because it cannot see the 13 unstable directions: the two obstructions are independent, and on high-dimensional chaos the observability wall binds first.

9.7 One axis: the top Lyapunov exponent

The linear theory of Sections 4 to 6 and the chaotic theory above are not two subjects but two signs of one quantity—the top Lyapunov exponent of the tangent cocycle Df, which for the LQG system is $\log \rho(A)$ and for the nonlinear systems is λ_1.

Proposition 9.10 (Stability margin and Lyapunov time are one dial). *Write the leading tangent multiplier as $r = e^{\lambda_1}$.*

1. (contracting side, $\lambda_1 < 0$) *The process is globally forecastable; the governing object is the* parameter *Fisher spectrum, with per-mode information $\Gamma_{jj} \asymp \mathrm{gain}_j^2/(1 - r_j^2)$ and $\lambda_{\max}(\Gamma) \asymp 1/(1-\rho)$ (Theorem 4.1).*
2. (chaotic side, $\lambda_1 > 0$) *The* state *is forecastable only within $\tau_\star \asymp \lambda_1^{-1} \log(1/\sigma_v)$ (Theorem 9.4).*
3. (the shared singularity) *As $\lambda_1 \to 0$ (equivalently $\rho \to 1$), both diverge at the* same *rate: since $1-r^2 = 1-e^{2\lambda_1} \sim -2\lambda_1$, the contracting-side information $1/(1-r^2) \sim 1/(2|\lambda_1|)$ and the chaotic-side horizon $\tau_\star \sim 1/\lambda_1$ are the one $1/|\lambda_1|$ singularity approached from the two sides of the unit circle.*

[figure not bundled with source]
tsfm_lqg/figs/fig_spectrum.pdf

Figure 2: Fisher spectrum of the TSFM on the 200-state LQG system. (a) the sloppy spectrum: ~ 15 decades with eight sensor-limited zeros; (b) gain-normalized per-mode information collapses onto the $1/(1-r^2)$ stability law (Theorem 4.1); (c) each small-eigenvalue eigenvector is a single mode.

Observability plays the dual role on each side: statically (unidentifiable parameter directions, Theorem 5.1) for $\lambda_1 < 0$, and dynamically (exponentially unpredictable state directions, Theorem 9.6) for $\lambda_1 > 0$.

Proof. Claims (1)–(2) restate Theorems 4.1 and 9.4. For (3), expand $1 - e^{2\lambda_1} = -2\lambda_1 + O(\lambda_1^2)$ as $\lambda_1 \to 0^-$, so the AR(1) information $\Gamma_{jj} \asymp 1/(1-r^2) \sim 1/(2|\lambda_1|)$; and $\tau_\star = \lambda_1^{-1}\log(1/\delta_0)$ from (15). Both scale as $|\lambda_1|^{-1}$ up to the slowly varying $\log(1/\delta_0)$ factor. ∎

The corpus (Table 5) instantiates the whole axis: LQG at $\lambda_1 = -0.105$ is the parameter-identification regime of Sections 2 to 6; Stuart–Landau and Kuramoto sit near the $\lambda_1 = 0$ singularity (a limit cycle and the synchronization edge), where both information and predictability are maximal but marginal; KS is weak low-dimensional chaos with a long $\tau_\star$; and Lorenz–96 at $\lambda_1 = +1.68$ is deep in the horizon-limited regime with a 13-dimensional unstable subspace that a low-m foundation model cannot fully observe. A single foundation model trained across all five is therefore doing parameter identification, near-critical tracking, and horizon-limited state forecasting at once—three problems the top Lyapunov exponent tells apart.

10 Numerical Study: A 200-State, 100-Observation Stable LQG System

We instantiate the theory on a deliberately large, stable, partially observed LQG system and a faithfully small CHRONOS-lite foundation model. The code is in `tsfm_lqg/` (system `lqg_system.py`, Fisher `fisher.py`, model `chronos_lite.py`, drivers `experiments*.py`).

The system. $K = 100$ modes give $n = 200$ latent states; $m = 100$ observations make C a 100×200 *fat* map (an observability ceiling by Theorem 5.1). Modal magnitudes are log-spaced in $[0.15, \rho]$ with baseline spectral radius $\rho(A) = 0.90$ (stability margin 0.10); process noise $\sigma_w = 1$, sensor noise $\sigma_v = 0.1$. Observation gains are tilted toward the slow, persistent modes (which therefore dominate each channel and are the most forecastable), and the **eight fastest modes are left unobserved** ($\gamma_j = 0$) to realize the sensor-limited case. The FIM (5) about $\theta = (r_1, \ldots, r_{100})$ is computed by the exact tangent Kalman filter; a Monte-Carlo score estimate agrees with the plug-in FIM to $\sim 10\%$.

The foundation model. CHRONOS-lite follows the CHRONOS recipe exactly: mean scaling, uniform quantization into $B = 128$ bins, and a small causal Transformer (decoder-only, $d_{\text{model}} = 64$, 2 layers, context 40) trained with the cross-entropy next-token loss; forecasts are autoregressive-sample or expected-value point forecasts, de-quantized and un-scaled. As CHRONOS prescribes, the model is *univariate*—one shared model across all 100 channels—so its access to each mode is limited by that mode's observability, making Theorem 5.1 operative.

10.1 The Fisher spectrum is sloppy and legible (Fig. 2)

Figure 2(a) shows the FIM eigenvalues spanning ~ 15 decades with exactly **eight** eigenvalues at machine zero—one per unobserved mode, confirming the $m < n$ null space of Theorem 5.1. Panel (b) plots the gain-normalized per-mode information $\Gamma_{jj}/\|c_j\|^2$ against the stability margin $1 - r_j^2$; it collapses onto the $1/(1-r^2)$ law of (6) with log–log correlation $\mathbf{0.97}$. Panel (c) shows the ten smallest-eigenvalue eigenvectors, each concentrated (weight ≈ 1) on a single fast or unobserved mode, confirming that the sloppy directions are individual physical modes.

[figure not bundled with source]
tsfm_lqg/figs/fig_dichotomy.pdf

Figure 3: The forecasting data–sensor dichotomy (Theorem 5.1). (a) more data cures excitation-limited but not sensor-limited directions; (b) adding channels that see the hidden modes lifts the null eigenvalue by eight orders of magnitude.

[figure not bundled with source]
tsfm_lqg/figs/fig_stability_fisher.pdf

Figure 4: Fisher properties versus stability (Theorem 4.1). (a) information grows toward the unit circle; (b) conditioning worsens; (c) $\lambda_{\max} \propto 1/(1-\rho)$, the stability-margin law.

10.2 Data versus sensor (Fig. 3)

Figure 3(a) plots the Cramér–Rao standard deviation $1/\sqrt{T\,\Gamma_{vv}}$ against series length T for three directions: a stiff (slow, observed) mode, an excitation-limited (fast, observed) mode, and a sensor-limited (unobserved) mode. The two observed directions fall as $T^{-1/2}$—*data cures them*, the stiff one far faster—while the unobserved direction is flat at its ceiling: no series length helps. Panel (b) shows that appending observation channels that see the eight hidden modes lifts the smallest eigenvalue on that subspace from $\approx 10^{-8}$ to ≈ 1, a jump of eight orders of magnitude: *only a new sensor cures the null*, exactly Theorem 5.1.

10.3 Stability writes the spectrum (Fig. 4)

Sweeping the spectral radius $\rho(A)$ from 0.40 to 0.98 (rescaling all modal magnitudes) traces Fig. 4. Panel (a): $\lambda_{\max}(\Gamma)$ and the dominant-mode information grow by more than a decade as $\rho \to 1$—information concentrates near the unit circle. Panel (b): the condition number worsens from $\sim 2\times 10^{14}$ to $\sim 5\times 10^{15}$—the sloppy spectrum widens. Panel (c): on log–log axes $\lambda_{\max}$ follows the predicted $1/(1-\rho)$ law of (7) (at $\rho = 0.98$, $\lambda_{\max} = 49.7 \approx 1/0.02$). Table 6 lists the numbers; they are the empirical content of Theorem 4.1.

Table 6: Fisher information versus stability margin (baseline system, exact tangent-filter FIM). Information ($\log\det\Gamma$, $\lambda_{\max}$) grows and conditioning (κ) worsens as $\rho(A) \to 1$; $\lambda_{\max}$ tracks $1/(1-\rho)$.

$\rho(A)$	0.40	0.55	0.68	0.78	0.86	0.92	0.96	0.98
$1-\rho$	0.60	0.45	0.32	0.22	0.14	0.08	0.04	0.02
$\log\det\Gamma$	−560.8	−553.6	−544.9	−536.1	−527.3	−519.0	−512.0	−507.7
$\lambda_{\max}(\Gamma)$	2.28	2.82	3.76	5.28	8.12	13.9	26.8	49.7
$1/(1-\rho)$	1.7	2.2	3.1	4.5	7.1	12.5	25	50
$\kappa(\Gamma)$ ($\times 10^{14}$)	2.3	2.8	3.8	5.3	8.1	13.9	26.7	49.7

10.4 The foundation model learns what the Fisher says (Fig. 5)

We train CHRONOS-lite on each stability level and measure one-step forecast R^2 (original units). Figure 5(a): the learned model's skill rises monotonically with $\rho(A)$, tracking the Kalman-optimal (all-channels) ceiling—more stable

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tsfm_lqg/figs/fig_tsfm_stability.pdf

Figure 5: CHRONOS-lite learning performance versus stability. (a) one-step forecast R^2 rises with the spectral radius toward the Kalman-optimal ceiling; (b) the same skill increases with the Fisher information $\log\det\Gamma$ (Proposition 6.1).

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tsfm_lqg/figs/fig_learning_curve.pdf

Figure 6: Learning curves and the Cramér–Rao law (Proposition 6.1). (a) forecast R^2 rises toward the Kalman ceiling with data; (b) the stiff and sloppy Fisher directions have vastly different CRB decay, so sloppy modes are learned last.

systems are more learnable, as Theorem 4.1 and Proposition 6.1 predict. Panel (b) plots the same skill against $\log\det\Gamma$: forecast performance is an increasing function of the Fisher information, the empirical content of Proposition 6.1. Table 7 gives the numbers: as $\rho(A)$ rises from 0.45 to 0.975 the learned model's one-step forecast R^2 climbs from 0.105 to 0.737, trailing the Kalman-optimal ceiling by a roughly constant gap—the univariate, finite-data penalty of Proposition 6.1—while $\log\det\Gamma$ rises in lockstep from −558.7 to −508.3.

Table 7: CHRONOS-lite forecast skill versus stability (trained per level; one-step R^2 on held-out series, original units). Learned skill rises with stability toward the Kalman-optimal ceiling and tracks the Fisher information $\log\det\Gamma$.

$\rho(A)$	0.45	0.60	0.72	0.82	0.90	0.95	0.975
TSFM R^2 (learned)	0.105	0.171	0.305	0.415	0.528	0.641	0.737
Kalman R^2 (ceiling)	0.124	0.224	0.351	0.492	0.623	0.720	0.800
$\log\det\Gamma$	−558.7	−550.5	−541.5	−531.7	−521.6	−513.5	−508.3

10.5 Learning curves and the Cramér–Rao law (Fig. 6)

Fixing $\rho = 0.90$ and varying the training budget, Fig. 6(a) shows forecast R^2 rising toward the Kalman ceiling as data grows—the learning curve of Proposition 6.1. Panel (b) plots the Cramér–Rao standard deviation $1/\sqrt{T\lambda}$ for the stiffest and the sloppiest identifiable Fisher directions; the two differ by the Fisher eigenvalue ratio (here $\sim 10^6$), so the sloppy modes need about a million times more data—*sloppy modes are learned last*, in exact proportion to their Fisher eigenvalue.

10.6 Summary of the numerical confirmations

Table 8 collects the quantitative checks. Every qualitative prediction of Sections 4 to 6 is borne out, and the two headline laws—$\Gamma_{jj} \propto 1/(1-r^2)$ and $\lambda_{\max} \propto 1/(1-\rho)$—hold to within numerical and Monte-Carlo error.

10.7 Cross-system forecasting and the Lyapunov horizon

We also train the single cross-system CHRONOS-lite of Table 4 on the full corpus of Section 8 and forecast each system's channels. Figure 7 shows representative steady-state dynamics of four environments; the oscillator systems (E2)–(E3)

Table 8: Numerical confirmations on the 200-state / 100-observation system.

Prediction	Theory	Measured
# null Fisher directions = # unobserved modes	8	8 (exact)
per-mode Fisher vs. stability margin	$\propto 1/(1-r^2)$	log–log corr 0.97
$\lambda_{\max}$ vs. spectral radius	$\propto 1/(1-\rho)$	Table 6
sensor cure of the null subspace	$0 \to> 0$	$10^{-8} \to 1.0$
FIM condition number as $\rho \to 1$	$\to \infty$	$2{\times}10^{14} \to 5{\times}10^{15}$
TSFM skill vs. stability / $\log\det\Gamma$	increasing	Fig. 5
sloppy vs. stiff learnability	$\propto \lambda$ ratio	$\sim 10^6$

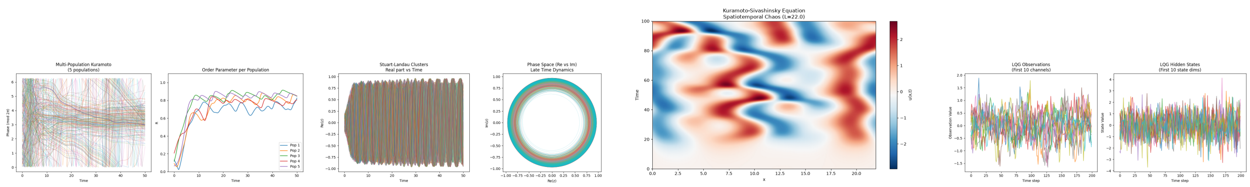

Figure 7: Representative dynamics of four corpus environments (left to right): multi-population Kuramoto (E2), cluster Stuart–Landau (E3), Kuramoto–Sivashinsky (E4), and the LQG observations (E1). The oscillator systems synchronize onto low-dimensional manifolds ($\lambda_1 \approx 0$); KS shows low-dimensional spatiotemporal chaos.

collapse onto low-dimensional synchronized manifolds (consistent with $\lambda_1 \approx 0$ and $d_B \approx 5$–7), while KS (E4) exhibits the sustained spatiotemporal chaos of a positive λ_1. Figure 8 contrasts the model's probabilistic forecasts on the stable LQG series and on the (near-critical) Kuramoto series. The behavior is exactly what Theorems 9.2 and 9.4 predict. On LQG ($\lambda_1 < 0$) the sample paths stay tight and skill persists over the whole horizon—there is no predictability wall, only the sloppy-mode accuracy floor of Proposition 6.1. On the oscillator/chaotic series the sample cloud fans out on the scale of the Lyapunov time $\tau_\star \asymp 1/\lambda_1$: for the fast, high-dimensional Lorenz–96 the useful horizon is a fraction of a time unit and a $C = 32$ univariate context is below the Takens window $2D_{\mathrm{KY}} + 1 \approx 55$ (Theorem 9.2, remark), so the model is both horizon- and observability-limited there; for the slow KS ($\tau_\star \approx 34$) it forecasts many steps ahead. The measured Lyapunov data of Table 5 thus predicts, system by system, how far the one foundation model can see.

11 Discussion

What the fusion buys. Reading a time-series foundation model as an autonomous world model imports a control-theoretic diagnostic kit into forecasting. The FIM $\mathcal{I}_{\theta\theta}$ is not an abstraction: its eigenvalues are the learnable rates of the system's modes, its small eigenvectors are the specific dynamics the model cannot resolve, and its null space is the observability ceiling of the sensor suite. A practitioner can, in principle, compute (or empirically estimate via per-sample gradient outer products [1, §12]) the model's Fisher spectrum and read off which series need longer histories, which need more channels, and which are hopeless until instrumentation changes—the same $\log\det$ and $\lambda_{\min}$ criteria that drive D- and E-optimal design.

Stability as the master dial. The stability–identifiability law (6) explains several empirical regularities of TSFMs at once: strongly autocorrelated (near-nonstationary) series are forecast with striking accuracy because their dominant modes carry enormous Fisher information; white-ish series are nearly unforecastable because they carry none; and the difficulty of a series is, to first order, its distance from the unit circle. It also warns that the same near-nonstationarity that makes a series learnable makes its FIM ill-conditioned, so second-order (natural-gradient / K-FAC) training and careful tokenization matter most exactly where forecasting is most rewarding.

From learnability to predictability. Crossing the unit circle changes the question a foundation model faces from *learnability* to *predictability* (Proposition 9.10). Below it ($\lambda_1 < 0$) enough data eventually identifies every observed mode; the only enemy is the sloppy spectrum. Above it ($\lambda_1 > 0$) the enemy is the dynamics itself: Theorem 9.4 caps every model's horizon at $\tau_\star \asymp \lambda_1^{-1}\log(1/\sigma_v)$, so the returns to scale—more parameters, more context, more

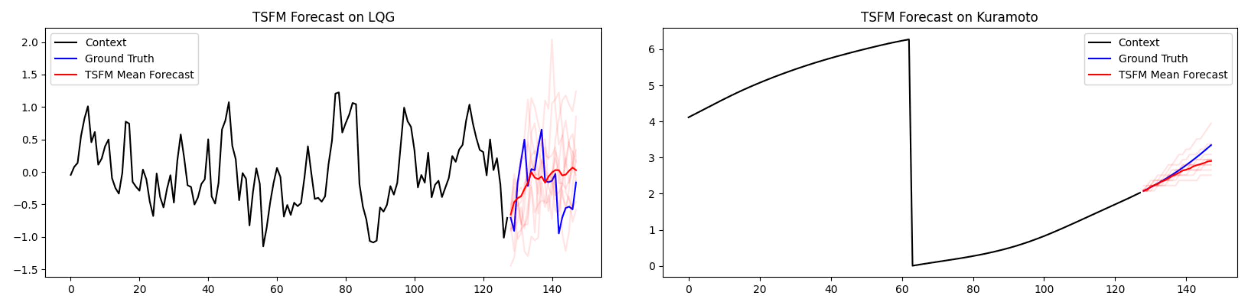

Figure 8: Cross-system CHRONOS-lite forecasts (context, ground truth, sample paths, and mean). Left: the stable LQG series (E1, $\lambda_1 < 0$)—no predictability wall, tight samples. Right: the Kuramoto series (E2, near the $\lambda_1 \approx 0$ synchronization edge)—the sample cloud spreads on the Lyapunov scale of Theorem 9.4.

corpus—collapse past $\tau_\star$, and the only levers are a quieter sensor (logarithmic gain) or a shorter sampling interval. This reframes "foundation-model" claims for physical time series: on chaotic data the ceiling is physical, not statistical, and benchmarks should report skill *in Lyapunov times* $h/\tau_\star$ rather than in raw steps, since $\tau_\star$ differs by two orders of magnitude across our corpus (Lorenz–96 vs. KS, Table 5). The complementary lever is Theorem 9.2: match the context length to the attractor's Takens window $2D_{\mathrm{KY}} + 1$, which for the low-dimensional oscillators is small but for Lorenz–96 exceeds the context we used—an architectural, not a data, deficiency.

Tokenization revisited. Proposition 2.2 places CHRONOS's "regression via classification" inside the information geometry: quantization is data processing that can only lose Fisher information, and the bin count B trades vocabulary size against the information retained about fine dynamics. The categorical head's advantage—arbitrary, multi-modal predictive distributions—is orthogonal to this and remains; the Fisher view simply prices the discretization.

Limitations. The exact spectral results are linear–Gaussian; nonlinear systems retain the structure through the extended/ensemble tangent filter but lose the closed-form $1/(1-r^2)$ law, which is why the chaotic theory (Section 9) is stated in terms of Lyapunov exponents rather than a closed-form spectrum. Several qualifications apply there. Theorem 9.2 is generic (Sauer–Yorke–Casdagli) and the constant $\mathrm{Lip}(F)$ can be large near tangencies; Theorem 9.4 is a minimax bound, tightened to a Bayes bound only under a posterior-excitation assumption; and the measured λ_1 for KS is specific to the repository's first-order IMEX integrator and step size, so its absolute value should be read as characterizing *this corpus* rather than the continuum PDE. We computed the full Lyapunov spectrum only for Lorenz–96; the oscillator/field dimensions in Table 5 marked † are design targets, and a full-spectrum $D_{\mathrm{KY}}/h_{\mathrm{KS}}$ for KS and the Kuramoto family is left to future work. Finally, we analyzed the one-step FIM and the one-step Lyapunov horizon; multi-step and rolling forecast risk propagate both through the dynamics, amplifing the role of the slow modes (contracting side) and of λ_1 (chaotic side) further.

12 Conclusion

A time-series foundation model, a world-action model, and a vision–language–action policy are three channels of one partially observed process on a shared approximate-information-state latent, and their learning problems are blocks of one Fisher information matrix. Deleting the action channel collapses the two-block theory to the single system-identification block, which is exactly what a forecaster optimizes; and for the linear–Gaussian instance that block's spectrum is written by the system's stability margins and observability. The slow modes near the unit circle carry the information and are learned first; the fast modes carry little and are learned last; the unobserved modes carry none and are never learned until a sensor is added. Beyond the unit circle the story continues by one letter: the tangent cocycle whose stable spectrum writes the parameter Fisher information also carries the Lyapunov exponents, and their sign is the master dial. On the contracting side ($\lambda_1 < 0$) the object is identifiability and the enemy is the sloppy spectrum; on the chaotic side ($\lambda_1 > 0$) the object is predictability and the enemy is the Lyapunov horizon $\tau_\star \asymp 1/\lambda_1$; and the two meet at $\lambda_1 = 0$, where the parameter-information blow-up $1/(1-r^2)$ and the predictability horizon $1/\lambda_1$ are one and the same $1/|\lambda_1|$ singularity. The context window is a delay-embedding information state of the attractor, so a modest model forecasts a high-dimensional field; but only the part of the unstable subspace it can observe, and only for a Lyapunov time. Stability and observability are the dials, predictability is the ceiling, and the same tangent-filter Fisher geometry that measures how well an agent can learn its world measures how well—and how far ahead—a foundation model can forecast a time series. Its small eigenvalues, and now its positive Lyapunov exponents, are the message.

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